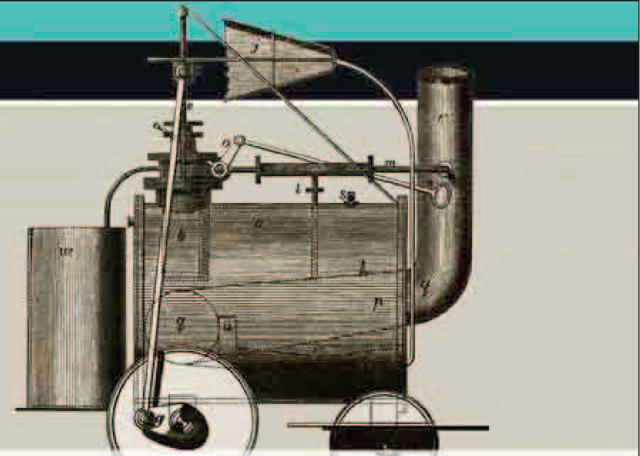
1801

> **“** THE **SUPERIORITY** OF THE **HIGH-PRESSURE ENGINE** WILL SERVE TO… **HONOR** THROUGH ALL TIMES… **THE NAME** OF **TREVITHICK. ”**

Michael Williams, Member of Parliament for West Cornwall, 1853



Richard Trevithick’s *Puffing Devil*—the world’s first passenger-carrying steam road locomotive—ran on the road, not on rails.

THOMAS YOUNG (1773–1829)

A practicing physician and polymath, Thomas Young contributed to understanding vision, light, mechanics, and energy, as well as language, music, and Egyptology. He helped in deciphering the Rosetta Stone. Elected fellow of the Royal Society in London at age 21, he became Professor of Natural Philosophy at the Royal Institution in 1801.

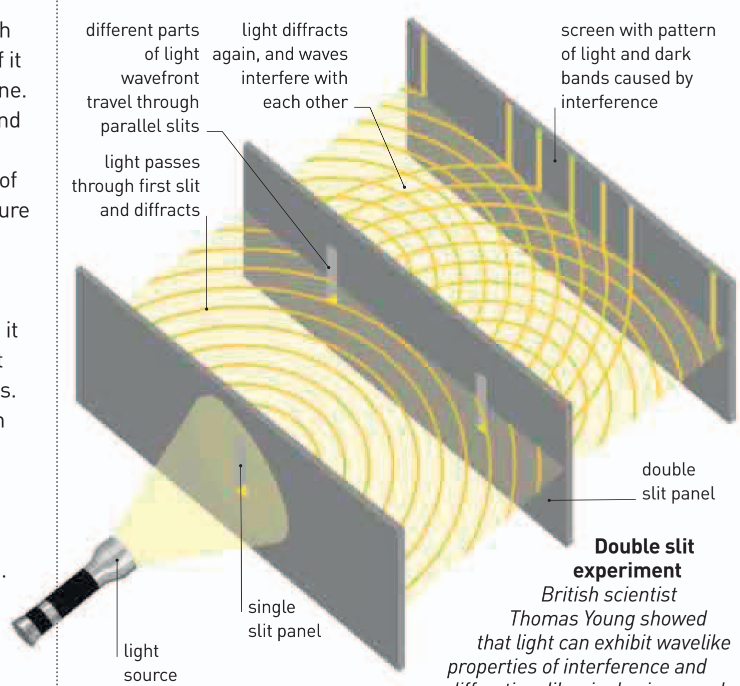
AT THE START OF THE YEAR, Italian astronomer **Giuseppe Piazzi** (1746–1826) **discovered Ceres**, a celestial body made of rock and ice that orbits the Sun every 1,679.819 days. Thought at the time to be a planet, it is now known to be the archetype of a class of smaller objects known as dwarf planets. Ceres is sometimes wrongly referred to as an asteroid because its orbit between Mars and Jupiter roughly coincides with that of a band of rocky rubble known as the asteroid belt.

The 19th century witnessed a growing understanding of atoms and molecules. In 1801, British chemist **John Dalton** developed his **law of partial pressures**, which stated that the pressure of a gas mixture is the sum of its individual components’ partial pressures—the pressure each component gas would exert if it occupied the same space alone. So, in a mixture of nitrogen and oxygen, the total pressure is equal to the partial pressure of oxygen plus the partial pressure of nitrogen.

Since the time of English physicist and mathematician **Isaac Newton** (see 1687–89), it had been widely believed that light was a stream of particles. In May 1801, British polymath Thomas Young (1773–1829) performed the **double slit experiment** (see right), which **demonstrated the wavelike properties of light**. It is now known that light sometimes behaves like waves and at other times like particles. The same year, German mathematician **Johann Georg von Soldner** (1776–1833) predicted how **light** traveling as a stream of particles would be **deflected by gravity** if it passed near the Sun. According to his calculation, a light ray would bend by 0.84 seconds of arc (a unit of angular measurement). More than 100 years later, German-American physicist Albert Einstein (1879–1955) would make a different prediction using his general theory of relativity (see pp.244–245).

British scientist **William Hyde Wollaston** (1766–1828) showed that electricity produced by friction (static electricity) and **galvanic electricity** produced by what is now referred to as a battery, are exactly the same.

French biologist **Jean-Baptiste Lamarck** (1744–1829) published his book *Système des Animaux Sans Vertèbres* (*System of Invertebrate Animals*), in which he coined the term **invertebrate** and developed a system of classification for this group of animals.



Double slit experiment
British scientist Thomas Young showed that light can exhibit wavelike properties of interference and diffraction, like ripples in a pond.

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MILES
THE DIAMETER OF **CERES**, THE **FIRST** KNOWN **DWARF PLANET**

The field of engineering was developing rapidly. Designed by Irish–American engineer **James Finley** (1756–1828), the **first suspension bridge hung from wrought-iron chains** was completed at Jacob’s Creek, Pennsylvania. It cost $600.

French inventor **Joseph-Marie Jacquard** (1752–1834) developed a system of controlling **textile looms** using punched cards. This was the **beginning of the era of programmable machines**.

On Christmas Eve, English engineer **Richard Trevithick** (1771–1833) successfully tested the **first successful steam road locomotive**. It carried several men up a hill in Camborne in Cornwall, UK, and traveled at approximately 4 mph (6.4 kmph).

January 1 Giuseppe Piazzi discovers the dwarf planet **Ceres**

John Dalton develops his **law of partial pressures**

Thomas Young conducts his **double slit experiment**, which shows that light is a wave

Johann Georg von Soldner uses Newtonian theory to calculate the **deflection of light**, as a stream of particles, by solar gravity

William Hyde Wollaston proves that static electricity and **galvanic electricity** are the same

German mathematician Carl Friedrich Gauss calculates the **orbit of Ceres**

Gauss publishes his influential textbook ***Arithmetical Investigations***

Jean-Baptiste Lamarck publishes ***System of Invertebrate Animals***

The **first iron-chain suspension bridge** opens in Pennsylvania

Joseph-Marie Jacquard develops the **Jacquard Loom**

November English chemist Charles Hatchett announces his discovery of the element **columbium**, later renamed **niobium**

December 24 Richard Trevithick tests his ***Puffing Devil***—the first successful steam road locomotive